1. Clinical Question & Core Hypothesis

Clinical Question

What molecular mechanisms determine whether immune structures translate into clinical benefit?

Working Hypothesis

Among BRCA1-mutant breast cancers in TCGA-BRCA, tumors that harbor evidence of homologous recombination (HR) restoration (operationalized by BRCA1 reversion mutations or functional HR readouts such as increased RAD51 activity and/or decreased BRCA1 promoter methylation leading to restored BRCA1 function) will show: (i) reduced HR deficiency signatures (lower COSMIC Signature 3 contribution and lower HRD score/LOH-based instability) and lower tumor mutational burden (TMB); (ii) diminished immune engagement evidenced by lower tumor-infiltrating lymphocytes (TIL density) and fewer TLS-like immune aggregates on pathology/imaging proxies; and (iii) decreased sensitivity to DNA-damaging therapies (PARP inhibitors and platinum agents) compared with BRCA1-mutant tumors without HR restoration. Additionally, imaging-derived features tied to immune infiltration (e.g., radiomics correlates of TLS/TIL presence) will be reduced in the HR-restored group. The central mechanistic expectation is that HR restoration mitigates genomic instability and neoantigen load, leading to a dampened anti-tumor immune milieu and attenuated response to DNA-damaging therapies. The study will test mediation pathways: BRCA1 mutation status → HR restoration status → HRD/Genomic instability metrics → TMB/neoantigen burden → TIL/TLS presence → therapy response, with imaging features serving as non-invasive proxies for immune milieu.

2. Why This Gene — Not Others

3. Clinically Testable Predictions

Prediction 1: The interaction between the imaging feature and anchor gene predicts clinical outcome.

Prediction 2: Spatial analysis will confirm the mechanistic link at the tissue level.

Prediction 3: Functional perturbation will establish causality.

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Novel mechanistic hypothesis derived from knowledge graph analysis.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions